

Slow Current Contributions to Long-lasting Spontaneous Burst Activity in Spiking Neural Network Model of Cultured Neuronal Networks

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Abstract—We studied spontaneous burst activity in a spiking neural network model with fast and slow synaptic currents. Simulations across multiple network sizes show that burst frequency remains nearly constant, demonstrating network-scale invariance consistent with that reported for cultured neuronal networks of comparable size. Although slow currents do not affect burst frequency, they substantially prolong burst duration. Dynamical analysis reveals that excitation and inhibition are balanced during resting states, while burst events emerge when excitatory synaptic currents became dominant. Fast AMPA-mediated currents trigger burst onset, whereas slow NMDA-mediated currents sustain activity over hundreds of milliseconds. Their interaction generates long-lasting spontaneous burst activity without altering invariant burst frequency.

Index Terms—cultured neuronal networks, spiking neural networks, spontaneous activity, burst firings, slow currents

I. INTRODUCTION

The human brain comprises billions of neurons interconnected through intricate synaptic networks. These neurons are organized into functional subpopulations, often referred to as “columns,” which are widely regarded as fundamental units of cortical information processing [1].

Recent experimental studies have highlighted the advantages of modular organization in cultured neuronal microcircuits, particularly in terms of the rich and complex intrinsic dynamics they exhibit [2], [3]. Such dynamics have been shown to enhance the computational capabilities of neuronal networks, especially within the framework of reservoir computing [4], [5].

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Computational (*in silico*) models have successfully reproduced input-driven neuronal activity under varying degrees of inter-modular connectivity and demonstrated the computational benefits of modular architectures [2], [4]. However, these models still fall short of fully capturing biological realism. In particular, they struggle to reproduce key features of spontaneous activity, such as network-size-invariant burst-event frequency that is invariant over a range of network sizes [6], as well as critical phenomena like neuronal avalanches, in which the size and duration of burst events follow scale-invariant distributions [3], [7]–[9].

To address the issue of network-size-invariant burst frequencies, we have previously incorporated synaptic weight scaling [10]–[13] into a *in silico* model that preserves the homeostasis [14]–[16] in its burst activities [17], [18]. Building on this approach, our previous study have investigated how lattice-based modular architectures shape spontaneous activity patterns in a biologically inspired spiking neuronal network (SNN) model [19]. Even though the SNN model enables to reproduce the scale-invariant burst event rate, the lengths of duration in burst events are much shorter than ones observed in cultured neuronal circuits. This incompatibility may deeply affect the information transmission among modules in networks.

Due to this, the SNN model should be modified to overcome the incompatibility. In the present study, we reconsider the SNN model with a single module and assume not only fast currents such as currents mediated through α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptors and γ -amino butyric acid A (GABA_A) receptors, but also slow currents such as currents mediated through N-methyl-D-aspartate (NMDA) receptors and GABA_B receptors in the

single-module SNN model.

II. IN SILICO MODEL OF CULTURED NEURONAL NETWORKS WITH SPIKING NEURON FRAMEWORK

We construct a spiking neuronal network model as a mathematical representation of cultured biological neuronal networks. The network consists of N neurons with an E/I ratio of 5 to 1. The N neurons are interconnected via synapses, and each neuron receives inputs from a total of $\sqrt{N}$ presynaptic neurons, whose ratio of E and I is also 5 to 1.

The subthreshold membrane dynamics are described by an extended leaky integrate-and-fire (LIF) model [20], which incorporates both relative refractoriness and Ca-dependent K currents:

$$C_m \dot{V} = g_L (V_{\text{rest}} - V) + I_{\text{ref}}(t) + I_{K(\text{Ca})}(t) + I(t),$$

where $\dot{} = d/dt$. Here, C_m , g_L , V_{rest} , and $V(t)$ denote the membrane capacitance, leak conductance, resting potential, and membrane potential, respectively. When $V(t)$ reaches the threshold V_{thre} , it is reset to V_{reset} . The additional currents are given by

$$I_{\text{ref}}(t) = g_{\text{ref}}(t) (V_{\text{reset}} - V), \\ I_{K(\text{Ca})}(t) = g_{K(\text{Ca})} c(t) (V_{K(\text{Ca})} - V),$$

where $V_{K(\text{Ca})}$ is the reversal potential of the Ca-dependent K current. The refractory conductance is defined as $g_{\text{ref}}(t) = \sum_{i=1}^3 g_{\text{ref}}^{(i)}(t)$, following a modified version of [20]:

$$\dot{g}_{\text{ref}}^{(i)} = -\frac{g_{\text{ref}}^{(i)}}{\tau_{\text{ref}}^{(i)}} + \sum_k \bar{g}_{\text{ref}}^{(i)} \delta(t - t_k),$$

where $\delta(t)$ is Dirac delta function, and t_k denotes the k -th spike time. The Ca concentration evolves as

$$\dot{c} = -\frac{c}{\tau_{\text{Ca}}} + \sum_k c_{\text{step}} \delta(t - t_k).$$

For I neurons, both $I_{\text{ref}}(t)$ and $I_{K(\text{Ca})}(t)$ are set to zero, reducing the model to the standard LIF dynamics. The total input current is given by $I(t) = I_{\text{syn}}(t) + I_{\text{bg}}(t)$, where $I_{\text{syn}}(t)$ and $I_{\text{bg}}(t)$ denote synaptic input and background noise, respectively. Both are modeled as $I_X(t) = I_E(t) + I_I(t)$ where $X \in \{\text{syn}, \text{bg}\}$.

For the background input ($X = \text{bg}$), the E and the I current have the form as the following:

$$I_Y(t) = g_Z(t) (V_Y - V), \quad Y \in \{E, I\},$$

where $Z = \text{AMPA}$ for $Y = E$ and $Z = \text{GABA}_A$ for $Y = I$. In each current, the conductance follows Ornstein-Uhlenbeck like dynamics [21]:

$$\dot{g}_Z = \frac{g_0 - g_Z}{\tau_Z} + \sigma_Z \eta(t), \quad Z \in \{\text{AMPA}, \text{GABA}_A\},$$

where $\eta(t)$ is Gaussian white noise with $\mathcal{N}(0, 1)$. The parameters g_0 and σ_Y are chosen such that isolated neurons fire at approximately 0.01 spk/s.

For synaptic input ($X = \text{syn}$), the currents involve both fast and slow current as

$$I_E(t) = g_{\text{AMPA}}(t) (V_E - V) + g_{\text{NMDA}}(t) B(V) (V_E - V), \\ I_I(t) = g_{\text{GABA}_A}(t) (V_I - V) + g_{\text{GABA}_B}(t) (V_I - V).$$

In these currents, the conductance evolves as

$$\dot{g}_Z = -\frac{g_Z}{\tau_Z} + \sum_{i,k} \bar{g}_Z^{(i)} x_i^-(t) u_i^+(t) \delta(t - (t_k^{(i)} + \Delta t^{(i)})), \quad (1)$$

where $Z \in \{\text{AMPA}, \text{NMDA}, \text{GABA}_A, \text{GABA}_B\}$, and $t_k^{(i)}$ and $\Delta t^{(i)}$ denote the k -th spike time of the i -th presynaptic neuron and the axonal delay, respectively. For $Z = \text{NMDA}$ and GABA_B , $\bar{g}_{\text{NMDA}}^{(i)} = 0.12 \bar{g}_{\text{AMPA}}^{(i)}$ and $\bar{g}_{\text{GABA}_B}^{(i)} = 0.1 \bar{g}_{\text{GABA}_A}^{(i)}$, respectively. In the E current, $B(V)$ is a function to mimic the NMDA-specific-voltage-dependent-gating feature, and is defined as

$$B(V) = \frac{1}{1 + 0.28 e^{-0.062V}}.$$

In Eq. (1), $x_i(t)$ and $u_i(t)$ are the short-term plasticity (STP) variables, where the former is the ratio of available neurotransmitter resources and the latter is the neurotransmitter release probability. The kinetics of these variables are defined as the followings:

$$\dot{x}_i = \frac{1 - x_i}{\tau_d^{(Y)}} - \sum_k x_i^-(t) u_i^+(t) \delta(t - (t_k^{(i)} + \Delta t^{(i)})), \quad (2) \\ \dot{u}_i = -\frac{u_i}{\tau_f^{(Y)}} + \sum_k U_i (1 - u_i^-(t)) \delta(t - (t_k^{(i)} + \Delta t^{(i)})), \quad (3)$$

where the parameter U_i determines the release probability for the first presynaptic spike arrival. The subscriptions $x_i^-(t)$ and $u_i^+(t)$ indicate $\lim_{t \rightarrow t_k^{(i)} + \Delta t^{(i)} - 0} x_i(t)$ and $\lim_{t \rightarrow t_k^{(i)} + \Delta t^{(i)} + 0} u_i(t)$ for each variable, respectively. This study keeps the parameter values of the SNN model employed in Ref. [18] except for τ_{Ca} , $g_{K(\text{Ca})}$, and $\bar{g}_Z^{(i)}$. In this study, we set $\tau_{\text{Ca}} = 2,550$ ms and $g_{K(\text{Ca})} = 17$ pS. The γ values are closer to the ideal value of 0.5 used in theoretical studies [11], [12], [22] and the estimated values in electrophysiological experiments [13] in comparison with our previous study [18].

From spike-raster data output from simulations of our SNN model, we identify bursts from the population firing rate, binned at 5 ms. Otsu's method sets the separating threshold directly from the rate histogram, splitting quiescent baseline from burst-time activity. The threshold is essentially size-independent (81–86 spikes/s). A burst is a maximal run of bins at or above threshold, with runs separated by less than 200 ms merged into one event, absorbing short reverberatory dips within an otherwise continuous burst.

III. RESULTS

A. Spontaneous long-lasting burst activity frequency is network-scale invariant

In all the network sizes, neurons exhibit burst spontaneous activity (Fig. 1(a)-(e)). In the network sizes of $N = 144$,

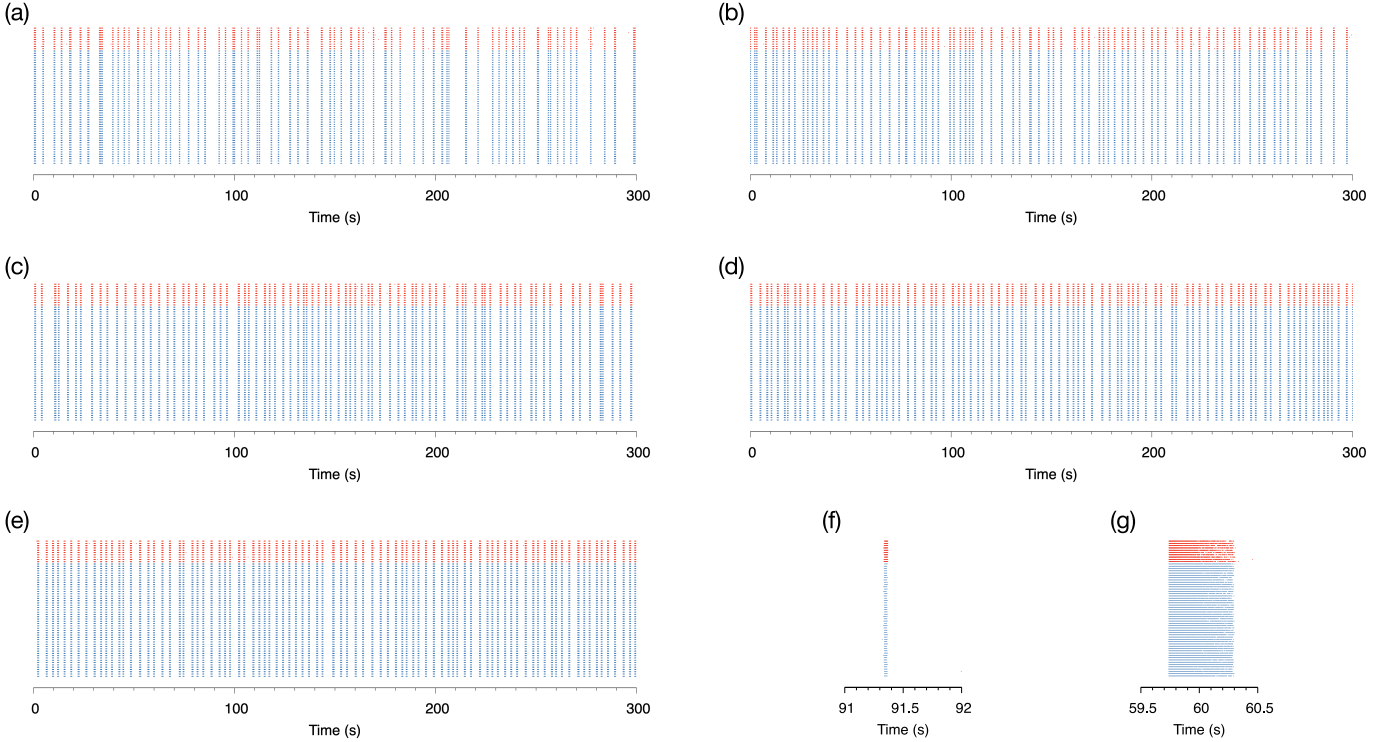


Fig. 1: Examples of spontaneous burst activities in the SNN model of cultured neuronal networks. (a)-(e) Raster plots for the network size N of 144, 324, 576, 900, and 1,296, respectively. (f) A burst event observed in the SNN in the absence of the slow currents [19]. The network size is the same as (e). (g) A burst event observed in the SNN in the presence of the slow currents. This is an enlargement of (e) in $t \in [59.5, 60.5]$ s. In all the panels, the blue and the red dots represent E and I neurons, respectively. Each panel shows spike trains of 50 E and 10 I neurons randomly chosen from individual assemblies of the SNN.

324, 576, 900, and 1,296, long time simulations estimate frequencies of burst event occurrences to be 0.255, 0.271, 0.273, 0.292, and 0.298 event/s (0.2533 ± 0.0059 , 0.2703 ± 0.0059 , 0.2731 ± 0.0049 , 0.2931 ± 0.0030 , and 0.2983 ± 0.0053 event/s for 20 different realizations), respectively, which indicates that burst event frequency of spontaneous neural activity is network-scale invariant. This result inherits from the previous SNN model without the slow currents in Ref. [19], and agrees with those reported for cultured neuronal networks in the corresponding size range [6]. Then, our SNN model is capable of reproducing the intrinsic homeostasis in spontaneous burst frequency whether the slow currents are present or absent.

The comparison of the absence and the presence of the slow currents tells us their role (Fig. 1(f)(g)). The absence of the slow currents allows burst activities in the model to last for tens milliseconds (Fig. 1(f)). In contrast, the slow currents makes lengths of burst activities longer (Fig. 1(g)). In our observations, the SNN with $N = 144, 324, 576, 900$, and 1,296 shows 397 ± 125 , 480 ± 70 , 586 ± 64 , 594 ± 64 , 634 ± 66 ms, respectively (mean $\pm$ SD).

Interestingly, our intensive parameter search for $\bar{g}_Z^{(i)}$ in Eq. (1) reveals that the SNN exhibits the spontaneous burst activities presented in Fig. 1 when $\gamma = 0.428, 0.485, 0.515$, and 0.624 for synaptic weights of the fast currents on $E \rightarrow E$,

$E \rightarrow I$, $I \rightarrow E$, and $I \rightarrow I$ connections, respectively, in $g = -(\alpha U)^{-1} \ln(1 - CK^{-\gamma}/A)$, which is obtained from the functions of $\text{PSP} = A(1 - e^{-\alpha U g})$ and $\text{PSP} = CK^{-\gamma}$, where K is the number of afferent connections on a neuron, A , α , C , and γ are the fitting parameters, and U are the STP parameter in Eq. (3). In these three functions, the units of PSP and g are in mV and nS. The values of A and α are estimated in the same manner in Ref. [18].

B. Interaction between fast and slow currents maintains spontaneous burst activity

In the previous section, we have confirmed the network-scale invariant frequency of spontaneous burst activity in the SNN model with the slow currents. In addition, it has been shown that the slow currents contribute spontaneous burst activities to being longer. In this section, we observe time series of the internal variables of an E neuron to understand the details of this mechanism (Fig. 2).

The subthreshold dynamics of the membrane potential $V(t)$ fluctuates around -58 mV (Fig. 2(a)) due to the zero-mean total current with the small fluctuation (Fig. 2(b)). Under this situation, the E and I background inputs are dominant in the total current and their mean values are almost same, which leads the mean total current to being zero (Fig. 2(g)(h)). This

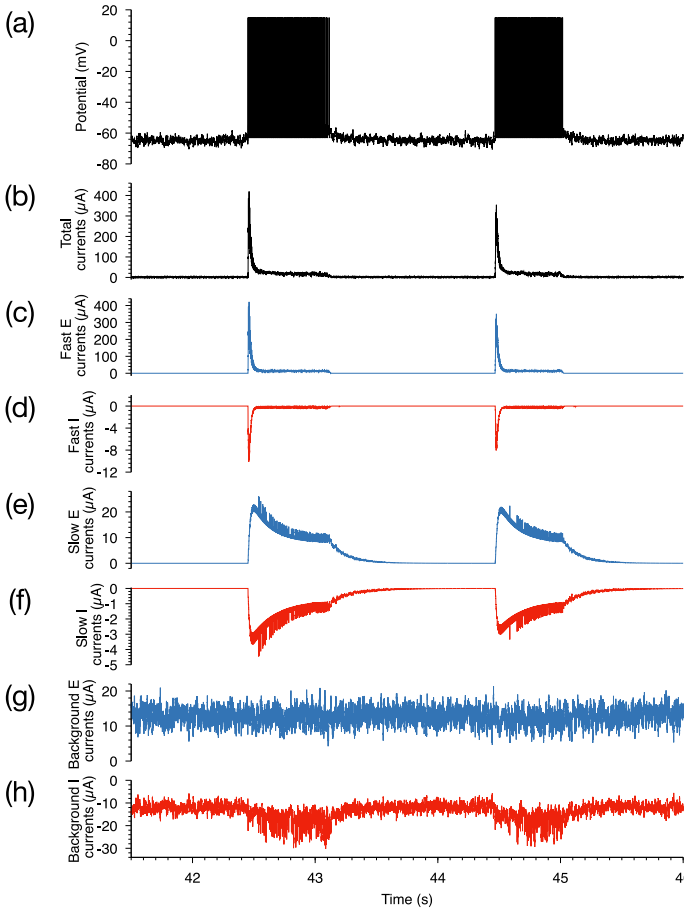


Fig. 2: Time series of the internal variables of an E neuron randomly selected from the SNN with $N = 1,296$. (a) Membrane potential $V(t)$, (b) total current $I(t)$, (c)(e) AMPA-mediated fast and NMDA-mediated slow E synaptic currents, (d)(f) GABA_A-mediated fast and GABA_B-mediated slow I synaptic currents, and (g)(h) E and I background inputs. The blue and red lines correspond to E and I currents, respectively.

cancellation of the E and I background inputs accomplishes precise E/I balance.

During the burst activities (Fig. 2(a)), E/I balance collapses, and the total current sustains positive values (Fig. 2(b)). Additionally, the dominance moves from the background inputs to the synaptic currents. Both fast and slow E currents take about 40 and 6.7 times as large as fast and slow I ones, respectively (Fig. 2(c)-(f)). At the beginnings of the burst firings, the total current shapes sharp peaks (Fig. 2(b)). The AMPA-mediated current defines the peaks and plays a role of triggers of the burst firings (Fig. 2(c)). This is consistent with our previous study because burst events occur when the SNN model does not include the slow currents (Fig. 1(f)). After the sharp peaks, the fast current does not go back to zero and maintains slightly positive values for about 550 ms. In these terms, the fast current is almost flat and works as a bias current.

In contrast, the peaks of the NMDA-mediated slow current is not large compared with those of the AMPA-mediated fast

current (Fig. 2(c)(e)). However, the peak widths of the slow current are quite wider than those of the fast current. The cooperation of the fast and slow E currents generates the almost flat total current lasting for about 550 ms after the sharp peaks in the total current (Fig. 2(b)). Consequently, this realizes the long-lasting spontaneous burst activities of neurons in the SNN network.

IV. CONCLUSIONS

In the present study, we have introduced slow currents mediated through NMDA and GABA_B receptors, and have demonstrated that, especially, the NMDA-mediated slow current contributes to extend the lengths of spontaneous burst events generated in the SNN model of cultured neuronal networks. Additionally, it has clarified that the AMPA-mediated fast and NMDA-mediated slow currents cooperates to accomplish long-lasting spontaneous burst firings of neurons.

We have succeeded to extend burst lengths in spontaneous neural activity from tens to hundreds milliseconds. Larger networks tend to more frequently exhibit longer bursts in cultured neuronal networks. Our SNN model reproduces not only homeostatic burst rate but also this trend.

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